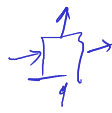


$$\frac{D\rho}{Dt} + \rho \frac{\partial v_j}{\partial x_j} = 0. \Rightarrow \rho \cdot \nabla \cdot \underline{v} = 0 \Rightarrow \nabla \cdot \underline{v} = 0$$

↓  
Incompressible



$$0 = \frac{D\rho}{Dt} = \frac{\partial \rho}{\partial x} \cdot v + \frac{\partial \rho}{\partial t} \Big|_{x=f(t)}$$